

Q1 - 26 July - Shift 2

The integral $\int \frac{\left(1 - \frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(1 + \frac{2}{\sqrt{3}} \sin 2x\right)} dx$ is equal to

(A) $\frac{1}{2} \log_e \left| \frac{\tan\left(\frac{x}{2} + \frac{\pi}{12}\right)}{\left(\frac{x}{2} + \frac{\pi}{6}\right)} \right| + C$

(B) $\frac{1}{2} \log_e \left| \frac{\tan\left(\frac{x}{2} + \frac{\pi}{6}\right)}{\left(\frac{x}{2} + \frac{\pi}{3}\right)} \right| + C$

(C) $\log_e \left| \frac{\tan\left(\frac{x}{2} + \frac{\pi}{6}\right)}{\tan\left(\frac{x}{2} + \frac{\pi}{12}\right)} \right| + C$

(D) $\frac{1}{2} \log_e \left| \frac{\tan\left(\frac{x}{2} - \frac{\pi}{12}\right)}{\tan\left(\frac{x}{2} - \frac{\pi}{6}\right)} \right| + C$

Space for your notes:

Q2 - 29 July - Shift 2

Questions

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For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$, if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

Space for your notes:

(A) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

(B) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

(C) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

(D) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

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Questions

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Answer Key

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Q1 (A)

Q2 (A)

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$$I = \int \frac{\left(1 - \frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(1 + \frac{2}{\sqrt{3}} \sin 2x\right)} dx$$

$$\frac{\sqrt{3}}{2} \int \frac{\left(1 - \frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(\frac{\sqrt{3}}{2} + \sin 2x\right)} dx$$

$$\int \frac{\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)(\cos x - \sin x)}{\sin 60^\circ + \sin 2x} dx$$

$$\int \frac{\left(\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \sin x\right)}{2 \sin\left(x + \frac{\pi}{6}\right) \cos\left(x - \frac{\pi}{6}\right)} dx$$

$$\int \frac{\left(\cos\left(x - \frac{\pi}{6}\right) - \sin\left(x + \frac{\pi}{6}\right)\right)}{2 \sin\left(x + \frac{\pi}{6}\right) \cos\left(x - \frac{\pi}{6}\right)} dx$$

$$\frac{1}{2} \left(\int \frac{dx}{\sin\left(x + \frac{\pi}{6}\right)} - \int \frac{dx}{\cos\left(x - \frac{\pi}{6}\right)} \right)$$

$$\frac{1}{2} \ln \left| \frac{\tan\left(\frac{x}{2} + \frac{\pi}{12}\right)}{\tan\left(\frac{x}{2} + \frac{\pi}{6}\right)} \right|$$

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Q2 (A)

$$I(x) = \int \sec^2 x \cdot \sin^{-2022} x \, dx - 2022 \int \sin^{-2022} x \, dx$$

$$= \tan x \cdot (\sin x)^{-2022} + \int (2022) \tan x \cdot (\sin x)^{-2023} \cos x \, dx$$

$$-2022 \int (\sin x)^{-2022} \, dx$$

$$I(x) = (\tan x) (\sin x)^{-2022} + C$$

$$\text{At } X = \pi/4, 2^{1011} = \left(\frac{1}{\sqrt{2}} \right)^{-2022} + C \therefore C = 0$$

$$\text{Hence } I(x) = \frac{\tan x}{(\sin x)^{2022}}$$

$$I(\pi/6) = \frac{1}{\sqrt{3} \left(\frac{1}{2} \right)^{2022}} = \frac{2^{2022}}{\sqrt{3}}$$

$$I(\pi/3) = \frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2} \right)^{2022}} = \frac{2^{2022}}{(\sqrt{3})^{2021}} = \frac{1}{3^{1010}} I\left(\frac{\pi}{6}\right)$$

$$3^{1010} I(\pi/3) = I(\pi/6)$$